

Supplementary tables for “Two-Phase Key Recovery of Forró: Bit Puncturing Meets Probabilistic Neutral Bits”. Table numbers here are local to this document.

Table 1: Per-component PNBs of the divide-and-conquer PNB phase for Forró 5/256. The output mask of the distinguisher is split into its single-bit components Γ_i . For each component i , P_i is the set of key bits that are neutral for that component alone, listed with the key bits numbered as in equation (1) of the paper; $|P_i|$ is its size, m_i is the number of key bits that still have to be guessed for the component, and ε_i is its conditional correlation.

i	Γ_i	$ P_i $	m_i	PNBs (P_i)	ε_i
1	$\Gamma_6^{(3)}[10]$	42	47	{5, 6, 7, 8, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 117, 118, 119, 120, 121, 133, 134, 135, 136, 137, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 236}	0.93
2	$\Gamma_{10}^{(3)}[0]$	55	34	{16, 17, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 133, 134, 135, 136, 137, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 232, 233, 236, 237, 238, 239, 240, 241, 242, 243}	1
3	$\Gamma_{10}^{(3)}[10]$	27	62	{12, 13, 14, 15, 16, 17, 126, 127, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 228, 229, 230, 231, 232, 233, 242, 243}	0.88
4	$\Gamma_{14}^{(3)}[0]$	50	39	{16, 17, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 232, 233, 236, 237, 238, 239, 240, 241, 242, 243}	1

Table 2: Per-component PNBs of the divide-and-conquer PNB phase for Forró 5.5/256. The output mask of the distinguisher is split into its single-bit components Γ_i . For each component i , P_i is the set of key bits that are neutral for that component alone, listed with the key bits numbered as in equation (1) of the paper; $|P_i|$ is its size, m_i is the number of key bits that still have to be guessed for the component, and ε_i is its conditional correlation.

i	Γ_i	$ P_i $	m_i	PNBs (P_i)	ε_i
1	$\Gamma_2^{(4)}[0]$	59	23	{24, 25, 26, 27, 28, 29, 30, 31, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 251, 252, 253, 254, 255}	1
2	$\Gamma_9^{(4)}[0]$	76	6	{8, 9, 10, 11, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 48, 49, 50, 59, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 96, 97, 98, 99, 100, 101, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 251, 252, 253, 254, 255}	1
3	$\Gamma_{14}^{(4)}[0]$	59	23	{24, 25, 26, 27, 28, 29, 30, 31, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 251, 252, 253, 254, 255}	1
4	$\Gamma_{14}^{(4)}[27]$	56	26	{21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 35, 36, 37, 38, 39, 40, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 234, 235, 236, 237, 238, 239, 240, 251}	0.94

Table 3: Per-component PNBs of the divide-and-conquer PNB phase for Forró 6/256. The output mask of the distinguisher is split into its single-bit components Γ_i . For each component i , P_i is the set of key bits that are neutral for that component alone, listed with the key bits numbered as in equation (1) of the paper; $|P_i|$ is its size, m_i is the number of key bits that still have to be guessed for the component, and ε_i is its conditional correlation.

i	Γ_i	$ P_i $	m_i	PNBs (P_i)	ε_i
1	$\Gamma_2^{(4)}[0]$	53	38	{3, 4, 8, 9, 10, 11, 12, 13, 41, 42, 43, 44, 48, 49, 50, 51, 80, 81, 82, 83, 96, 120, 121, 122, 136, 137, 152, 153, 154, 155, 156, 157, 158, 160, 161, 162, 163, 164, 165, 229, 230, 231, 241, 242, 243, 244, 245, 246, 247, 252, 253, 254, 255}	0.92
2	$\Gamma_9^{(4)}[0]$	53	38	{8, 9, 10, 11, 12, 13, 14, 15, 16, 48, 49, 59, 60, 61, 62, 63, 131, 132, 133, 134, 135, 136, 137, 152, 153, 154, 155, 156, 157, 158, 160, 161, 162, 163, 164, 165, 166, 167, 168, 229, 230, 231, 241, 242, 243, 244, 245, 246, 247, 252, 253, 254, 255}	0.98
3	$\Gamma_{14}^{(4)}[0]$	52	39	{3, 4, 8, 9, 10, 11, 12, 13, 14, 41, 42, 43, 44, 48, 49, 50, 51, 52, 80, 81, 82, 91, 120, 121, 122, 136, 137, 152, 153, 154, 155, 156, 157, 158, 160, 161, 162, 163, 164, 165, 166, 241, 242, 243, 244, 245, 246, 247, 252, 253, 254, 255}	0.85
4	$\Gamma_{14}^{(4)}[27]$	42	49	{3, 4, 8, 9, 10, 41, 42, 43, 44, 48, 49, 59, 60, 78, 79, 80, 96, 118, 119, 120, 131, 132, 133, 152, 153, 154, 155, 156, 158, 160, 161, 162, 229, 230, 231, 241, 242, 243, 252, 253, 254, 255}	0.40

Table 4: Trail enumeration for f_1 in the 5.5-round attack. The enumeration starts from the target bit $v_{14}^{[18]}$ [27] and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and *green* bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j ; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_1^2 .

Subround	Active	#coeffs	ρ^2
18	$v_{14}^{[18]}$ [27]	1	1
19	$v_{14}^{[18]}$ [27] $\leftarrow \{t_{14}$ [27, 26–19], t_1 [27, 26–19] $\}$ [add]	510	$2^{-0.01}$
	t_1 [27, 26–19] $\leftarrow \{v_1^{[19]}$ [27, 26–19], t_2 [27, 26–19] $\}$ [xor]	510	$2^{-0.01}$
	t_{14} [27, 26–19] $\leftarrow \{v_{14}^{[19]}$ [22, 21–17], $v_1^{[19]}$ [26–19] $\}$ [add]	31048	$2^{-0.03}$
	t_2 [27, 26–19] $\leftarrow \{v_2^{[19]}$ [31–29, 3, 2–0], $v_6^{[19]}$ [27, 26–20] $\}$ [add]	317376	$2^{-0.09}$
20	$v_2^{[19]}$ [31–29, 3, 2–0] $\leftarrow \{v_2^{[20]}$ [31–29, 3, 2–0], t_3 [31–29, 3, 2–0] $\}$ [xor]	317376	$2^{-0.09}$
	t_3 [31–29, 3, 2–0] $\leftarrow \{v_3^{[20]}$ [11, 10–5], $v_7^{[20]}$ [31, 30, 3, 2, 1] $\}$ [add]	357416	$2^{-0.76}$
21	$v_3^{[20]}$ [11, 10–5] $\leftarrow \{v_3^{[21]}$ [11, 10–5], t_0 [11, 10–5] $\}$ [xor]	357416	$2^{-0.76}$
	t_0 [11, 10–5] $\leftarrow \{v_0^{[21]}$ [19, 18–13], $v_5^{[21]}$ [11, 10–8] $\}$ [add]	1288208	$2^{-1.13}$
22	$v_6^{[21]}$ [27, 26–20] $\leftarrow \{t_6$ [5, 4–2], t_{11} [27, 26–21] $\}$ [add]	2910336	$2^{-1.22}$
	$v_1^{[21]}$ [26–19] $\leftarrow \{t_1$ [26–21], t_6 [26–24, 5, 4–2] $\}$ [add]	4511424	$2^{-1.43}$
	$v_0^{[21]}$ [19, 18–13] $\leftarrow \{v_0^{[22]}$ [19, 18–13], t_1 [26–21, 19, 18–13] $\}$ [xor]	4511424	$2^{-1.43}$
	t_{11} [27, 26–21] $\leftarrow \{v_{11}^{[22]}$ [27, 26–21], $v_{12}^{[22]}$ [27, 26–24] $\}$ [add]	8384832	$2^{-1.53}$
	t_6 [26–24, 5, 4–2] $\leftarrow \{v_6^{[22]}$ [26–24, 5, 4–2], $v_{11}^{[22]}$ [27, 26–21, 5, 4–2] $\}$ [xor]	8384832	$2^{-1.53}$
	t_1 [26–21, 19, 18–13] $\leftarrow \{v_1^{[22]}$ [31–29, 27, 26–21, 2–0], $v_6^{[22]}$ [25–23, 19, 18–16, 5, 4–2] $\}$ [add]	20479104	$2^{-2.13}$
	$v_{14}^{[22]}$ [22, 21–17] = $v_{14}^{[19]}$, $v_5^{[22]}$ [11, 10–8] = $v_5^{[21]}$, $v_3^{[22]}$ [11, 10–7] = $v_3^{[21]}$ [rename]	20479104	$2^{-2.13}$
end	$v_{14}^{[22]}$ [22, 21–17], $v_{12}^{[22]}$ [27, 26–24], $v_7^{[22]}$ [31, 30, 3, 2, 1], $v_6^{[22]}$ [25–23, 19, 18–16, 5, 4–2], $v_5^{[22]}$ [11, 10–8], $v_{11}^{[22]}$ [27, 26–21, 5, 4–2] $\leftarrow \{\bar{z}_{11}$ [27, 26–23, 5, 4, 3], k_{11} [27, 26–21, 5, 4–0] $\}$, $v_3^{[22]}$ [11, 10–7] $\leftarrow \{\bar{z}_3$ [11, 10–8], k_3 [11, 10–6] $\}$, $v_2^{[22]}$ [31, 3, 2–0] $\leftarrow \{\bar{z}_2$ [31–28, 3, 2–0], k_2 [31–29, 3, 2–0] $\}$, $v_1^{[22]}$ [31–29, 27, 26–21, 2–0] $\leftarrow \{\bar{z}_1$ [31–28, 27, 26–23, 2–0], k_1 [31–28, 27, 26–19, 2–0] $\}$, $v_0^{[22]}$ [19, 18–13] $\leftarrow \{\bar{z}_0$ [19, 18–15], k_0 [19, 18–12] $\}$	$\leq 2^{55+44}$	$2^{-2.96}$

Table 5: Trail enumeration for f_2 in the 5.5-round attack. The enumeration starts from the target bit $v_0^{[21]}[8]$ and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and *green* bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j ; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_2^2 .

Subround	Active	#coeffs	ρ^2
21	$v_0^{[21]}[8]$	1	1
22	$v_0^{[21]}[8] \leftarrow \{v_0^{[22]}[8], t_1[8]\}$ [xor]	1	1
	$t_1[8] \leftarrow \{v_1^{[22]}[16, 15-8], v_6^{[22]}[8, 7-0]\}$ [add]	766	1
end	$v_6^{[22]}[8, 7-0],$ $v_1^{[22]}[16, 15-8] \leftarrow \{\bar{z}_1[16, 15-6], k_1[16, 15-9]\},$ $v_0^{[22]}[8] \leftarrow \{\bar{z}_0[8, 7-0], k_0[8, 7-0]\}$	$\leq 2^{26+15}$	$2^{-0.01}$

Table 6: Trail enumeration for f_3 in the 5.5-round attack. The enumeration starts from the target bit $v_{11}^{[22]}[10]$ and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and *green* bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j ; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_3^2 .

Subround	Active	#coeffs	ρ^2
22	$v_{11}^{[22]}[10]$	1	1
end	$v_{11}^{[22]}[10] \leftarrow \{\bar{z}_{11}[10, 9-0], k_{11}[10, 9-0]\}$	3070	1